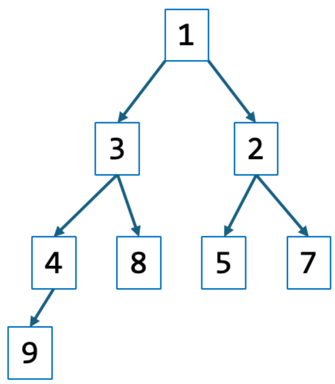
CMPT 225-D100 Quiz 3, Summer 2026

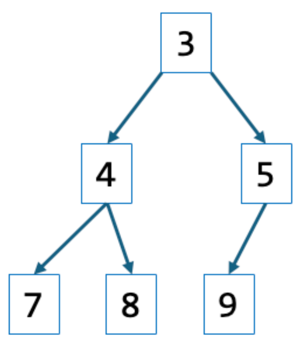
Duration	30 minutes for the entire exam.
Coding Questions	For coding questions, use C++-like pseudocode. This means that small coding errors will usually be overlooked. However, it must be clear that the code could easily be converted to real C++.
Aids allowed	Pencil or pen, and eraser. No notes, no papers, no books, no computers, no calculators, no cheat sheets, etc.
During the exam	Raise your hand if you would like to speak with a proctor. To be fair to all students, we don’t answer questions about the exam. Choose the answer based on your best understanding of the question. Re-reading the question may help.

1) State the **heap order property** (for a min-heap). For every node v other than the root, the key of v is greater than, or equal to, the key of its parent.	
2) What, exactly, is the height of a **heap** with 100 nodes? If there is a range of possible heights, then clearly state the range. Recall that the height of a leaf is 0.	height = 6 **1 mark** $\lfloor \log 100 \rfloor$ is okay (-0.5 if missing floor, or if use ceiling)

3) Draw the min-heap that results from inserting these keys into an initially empty heap:
9, 5, 4, 2, 8, 1, 7, 3



4) Starting with the final heap from 3) with all 8 elements inserted, draw the min-heap that results after calling the remove min function **twice** on the heap (so the final heap has 6 elements).



5) Suppose you have an initially empty hash table of size $n = 10$.

a) Using **linear probing** (as in the textbook) with the hash function $h(k) = k \; mod \; 10$, insert these numbers in this order: **31**, **29**, **48**, **12**, **18**, **59**

0	1	2	3	4	5	6	7	8	9
18	31	12	59					48	29

b) Using **quadratic probing** (as in the textbook) with the hash function $h(k) = k \; mod \; 10$, insert these numbers in this order: **4**, **6**, **14**, **54**, **84**, **5**

0	1	2	3	4	5	6	7	8	9
			84	4	14	6		54	5

c) What is the **load factor** of the completed hash table in part b)?

load factor $\lambda = \frac{6}{10} = 0.6$